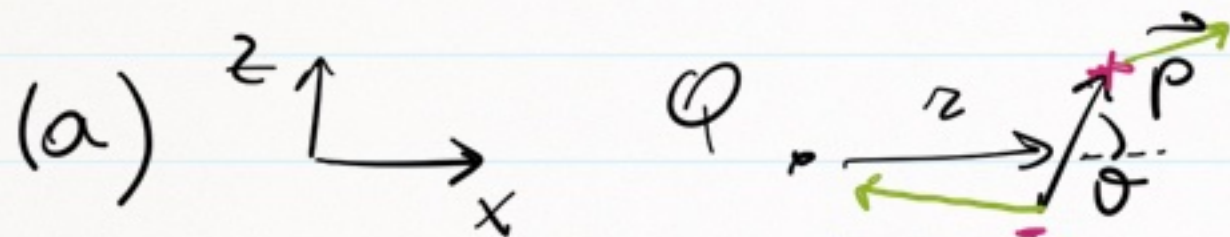
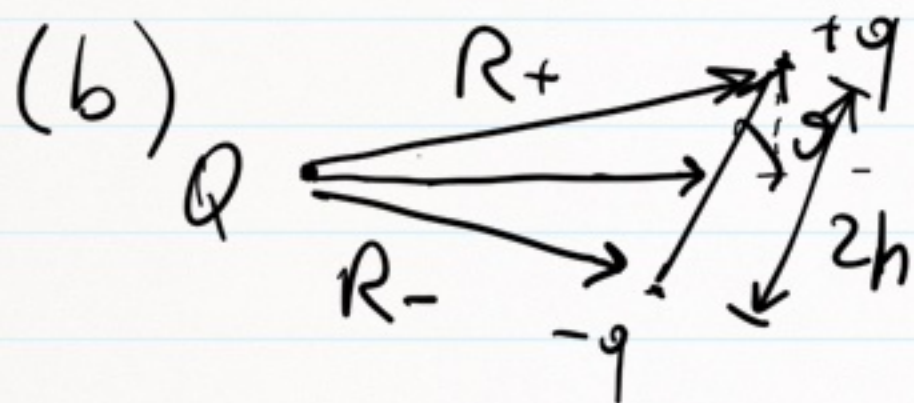


21/10/20

④ Ex. 1



$$\left\{ \begin{array}{l} E_x = \dots \\ E_y = 0 \\ E_z = \dots \end{array} \right.$$



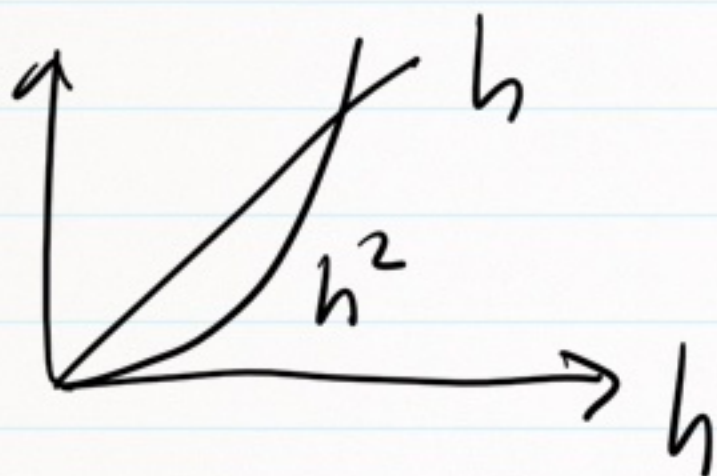
$$R_{\pm}^2 = (z \mp h \cos \theta)^2 + (h \sin \theta)^2$$

$$F_{\pm} = \pm \frac{q Q}{4\pi\epsilon_0 R_{\pm}^2}$$

$$F = \lim_{h \rightarrow 0} F_+ + F_- = \frac{q Q}{4\pi\epsilon_0} \left(\frac{1}{r^2 + 2zh \cos \theta + h^2} - \frac{1}{r^2 - 2zh \cos \theta + h^2} \right)$$

$$(p = 2qh \rightarrow q = p/2h)$$

$$\lim_{h \rightarrow 0} \frac{Qp}{4\pi\epsilon_0} \cdot \frac{1}{2h} \left(\frac{r^2 - 2rh\cos\theta - r^2 - 2rh\cos\theta + \cancel{h^2} - \cancel{h^2}}{r^4 - \cancel{4r^2h^2\cos^2\theta}} \right) =$$



h domine sur h^2

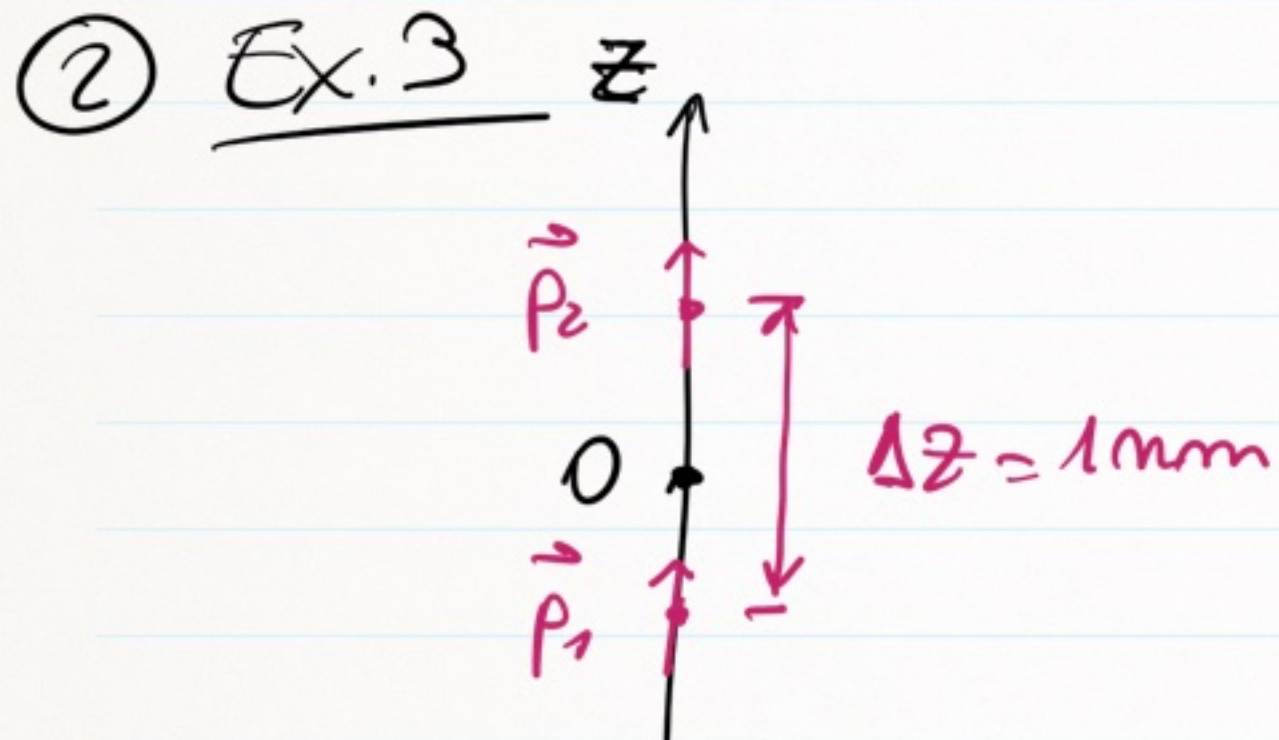
$$= - \frac{Qp}{4\pi\epsilon_0} \frac{4r\cos\theta}{2r^4} = - \frac{Qp}{2\pi\epsilon_0 r^3} \cdot \cos\theta$$

$$(c) \quad u_e = - \vec{E} \cdot \vec{r} = - \frac{Q}{4\pi\epsilon_0 r^2} r \cos\theta$$

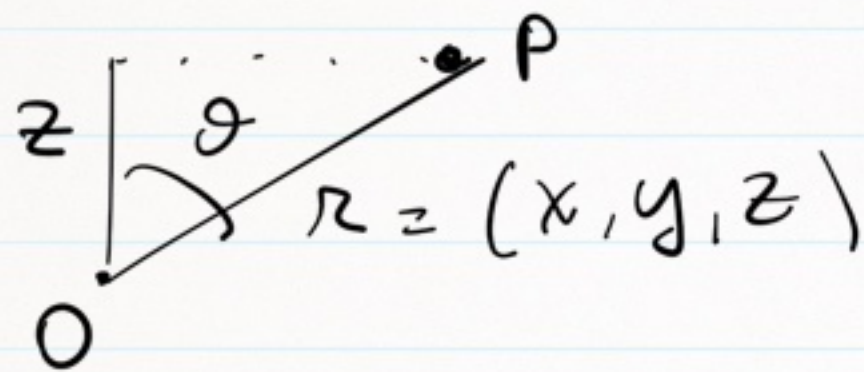
$$F = - \nabla u_e = - \frac{2Q}{4\pi\epsilon_0 r^3} r \cos\theta$$

$$M = - \frac{\partial u_e}{\partial \theta} = - \frac{Qr}{4\pi\epsilon_0 r^3} \sin\theta$$

ROT. DIPOLO



$$V_{P_i}' = \frac{p_i r \cos\theta}{4\pi\epsilon_0 r^3}$$



$$r = \sqrt{x^2 + y^2 + z^2}$$

$$r \cos \theta = z$$

$$V_p = \frac{p}{4\pi\epsilon_0} \cdot \frac{z}{(x^2 + y^2 + z^2)^{3/2}}$$

$$E_z = - \frac{\partial V_p}{\partial z} = - \frac{p}{4\pi\epsilon_0} \left[\frac{1}{(x^2 + y^2 + z^2)^{3/2}} + \right.$$

$$\left. - \frac{3z^2}{(x^2 + y^2 + z^2)^{5/2}} \right] =$$

$$= \frac{1}{4\pi\epsilon_0} \left[- \frac{p}{z^3} + \frac{3 \vec{p} \cdot \vec{r}}{z^5} z \right] =$$

$$= \frac{1}{4\pi\epsilon_0} \left[\left(\frac{3 \vec{p} \cdot \vec{r}}{r^5} \right) \vec{r} - \frac{\vec{p}}{r^3} \right]$$

$$\xrightarrow{\theta=0} \frac{1}{4\pi\epsilon_0} \frac{2p}{\Delta z^3}$$

$$U_{e, 2 \leftarrow 1} = -p_2 E_1 = \frac{2p_1 p_2}{4\pi\epsilon_0 \Delta z^3}$$

$$F_{e, 2 \leftarrow 1} = - \frac{\partial U_{e, 2 \leftarrow 1}}{\partial z} = - \frac{3p_1 p_2}{2\pi\epsilon_0 \Delta z^4}$$

$$F_{e,2 \leftarrow 1} = - \frac{3 (6.3)^2 \cdot 10^{-60} \text{ C}^2 \cdot \text{m}^2}{2\pi \cdot 8.85 \cdot 10^{-12} \frac{\text{C}^2}{\text{Nm}^2} \cdot 10^{-36} \text{ m}^4} \approx$$

$$\approx 2.12 \cdot 10^{-12} \text{ N} = 2.12 \text{ pN}$$

③ Ex. 4

1:	$q_1 = -q_0$	$\vec{r}_1 \equiv (d, 0, 0)$
2:	$q_2 = +q_0$	$\vec{r}_2 \equiv (0, -d, 0)$
3:	$q_3 = -q_0$	$\vec{r}_3 \equiv (-d, 0, 0)$
4:	$q_4 = +q_0$	$\vec{r}_4 \equiv (0, d, 0)$

MONOPOL: $\sum_{i=1}^4 q_i = 0$

DIPOLLO : $\vec{p} = \sum_{i=1}^4 q_i \vec{r}_i$

$$p_x = \sum_{i=1}^4 q_i x_i = -q_0 d - q_0 (-d) = 0$$

$$p_y = \sum_{i=1}^4 q_i y_i = -q_0 d + q_0 d = 0$$

$$p_z = 0$$

QUADRUPOLO : $Q = \sum_{i=1}^4 q_i \left[\frac{3}{2} (\vec{r}_i \cdot \hat{r})^2 - \frac{(r_i)^2}{2} \right] \leftarrow$

$$= \sum_{i=1}^4 q_i r_i^2 \frac{3\cos^2\theta - 1}{2}$$

$$V = \frac{1}{4\pi\epsilon_0} \left[\sum_i \frac{q_i}{r} + \frac{\vec{p} \cdot \hat{r}}{r^2} + \underbrace{\frac{1}{r^3} Q}_{\text{quad}} + \dots \right]^2$$

$$\hat{n} \equiv \hat{y}$$

$$Q = -q_0 \left[\frac{3}{2} (\theta)^2 - \frac{d^2}{2} \right] +$$

$$q_0 \left[\frac{3}{2} (-d)^2 - \frac{(-d)^2}{2} \right] +$$

$$- q_0 \left[\frac{3}{2} (\theta)^2 - \frac{(-d)^2}{2} \right] +$$

$$q_0 \left[\frac{3}{2} d^2 - \frac{d^2}{2} \right] =$$

$$= q_0 \frac{d^2}{2} + q_0 d^2 + q_0 \frac{d^2}{2} + q_0 d^2 = \underline{3q_0 d^2}$$

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{D^3} = \frac{3q_0 d^2}{4\pi\epsilon_0 D^3}$$

④ Ex. 5 $\underline{d\tau} = r^2 \sin\theta dr d\theta d\varphi$

MONOPOLE : $q = \int \rho d\tau = KR \int \left[\frac{1}{r^2} (R-2r) \sin\theta \right] d\tau$

in r , $\int_0^R \frac{1}{r^2} (R-2r) r^2 dr = [Rr - r^2]_0^R = 0$

$$V = \frac{1}{4\pi\epsilon_0} \left(\underbrace{\frac{q}{r}} + \underbrace{\frac{\vec{p} \cdot \vec{r}}{r^2}} + \frac{1}{r^3} \int r'^2 \cdot \frac{3\cos^2\theta - 1}{2} \rho dV_{+...} \right)$$

per DIPOLO : $\vec{p} \cdot \hat{r} = \int r \cos \theta \rho d\tau$

in r , $\int_0^R \frac{1}{r} (R-2r) r^2 dr = \left[\frac{Rr^2}{2} - \frac{2r^3}{3} \right]_0^R =$

$$= \frac{R^3}{2} - \frac{2R^3}{3} = -\frac{R^3}{6}$$

in θ , $\int_0^\pi \cos \theta \sin^2 \theta d\theta = \int_0^\pi t^2 dt = 0$
 $\left(\begin{array}{l} t = \sin \theta \\ dt = \cos \theta d\theta \end{array} \right)$

per QUADRUPOLO :

$$\int r^2 \cdot \frac{3 \cos^2 \theta - 1}{2} \rho d\tau$$

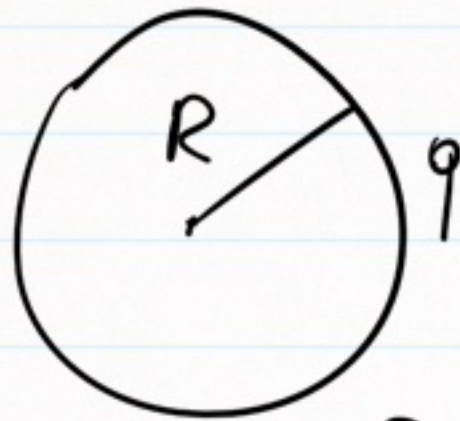
$$\text{in } r, \int_0^R r^2 (R - 2r) dr = \dots = -\frac{R^4}{6}$$

$$\text{in } \theta, \int_0^\pi \underbrace{(3 \cos^2 \theta - 1)}_{2 - 3 \sin^2 \theta} \sin^2 \theta d\theta = \dots = \pi - \frac{9\pi}{8}$$

$$\text{in } \varphi, \int_0^{2\pi} d\varphi = 2\pi$$

$$V(z) = \frac{4\pi^2 R^3}{48} \cdot \frac{1}{4\pi\epsilon_0} \frac{1}{z^3}$$

⑤ Ex. 6



$$R = 10 \text{ cm}$$

$$q = 10 \text{ nC}$$

$$(a) \sigma = \frac{q}{4\pi R^2} = \frac{10^{-8} \text{ C}}{4\pi \cdot 10^{-1} \text{ m}^2} = 7.96 \cdot 10^{-9} \frac{\text{C}}{\text{m}^2}$$

$$(b) E = \frac{\sigma}{\epsilon_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} =$$

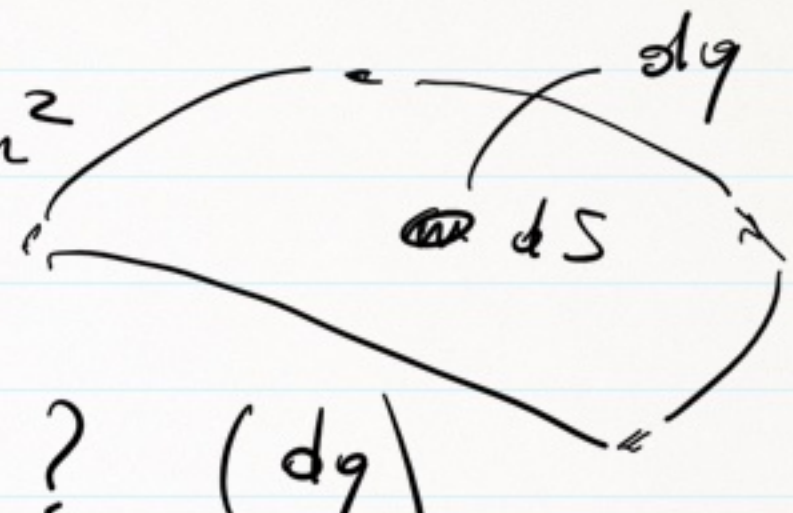
$$= \frac{7.96 \cdot 10^{-9} \text{ C/m}^2}{8.85 \cdot 10^{-12} \frac{\text{C}^2}{\text{Nm}^2}} = 899.44 \frac{\text{N}}{\text{C}}$$

(c) $F = p_e \cdot \tilde{S}$, $\tilde{S} = 1 \text{ mm}^2$

1 MOODO

$$p_e = \frac{|dF|}{dS}$$

$dF = ?$ $\left(\frac{dq}{\sigma dS} \right)$



$$dF = dq \cdot \underbrace{E_{\text{lon}}}_{\frac{\sigma}{2\epsilon_0}} = \sigma dS \frac{\sigma}{2\epsilon_0} = \frac{\sigma^2}{2\epsilon_0} dS$$

$$p_e = \frac{\sigma^2}{2\epsilon_0} = \frac{1}{2} \frac{(\sigma)}{\epsilon_0} \sigma = \frac{1}{2} \sigma E$$

$$F = p_e \cdot \tilde{S} = \frac{1}{2} \sigma E \tilde{S}$$

sostituire σ, E
dalla pagina precedente

$$F = \frac{1}{2 \cdot \pi^2 \cdot \epsilon_0} \cdot \frac{q^2}{R^4} \cdot \tilde{S} = \dots = 3.58 \cdot 10^{-10} \text{ N}$$

N.B: ad esercitazione avevo erroneamente sostituito $S = 4\pi R^2$,
ma non corrispondeva alla richiesta del punto (c)

1 ALTRO MODO

$$dF = dq E = dq \cdot \frac{\sigma}{\epsilon_0} = dq \frac{q}{S \epsilon_0}$$

$$F = \int \frac{q dq}{S \epsilon_0} = \frac{1}{2} \frac{q^2}{S \epsilon_0} = \frac{1}{2} \cdot \frac{1}{4\pi R^2} \frac{q^2}{\epsilon_0}$$

$$P_e = \frac{F}{S} = \frac{\sigma^2}{2 \epsilon_0}$$

$$(d) V = \frac{1}{4\pi\epsilon_0} \oint_S \frac{\sigma dS}{r} = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$

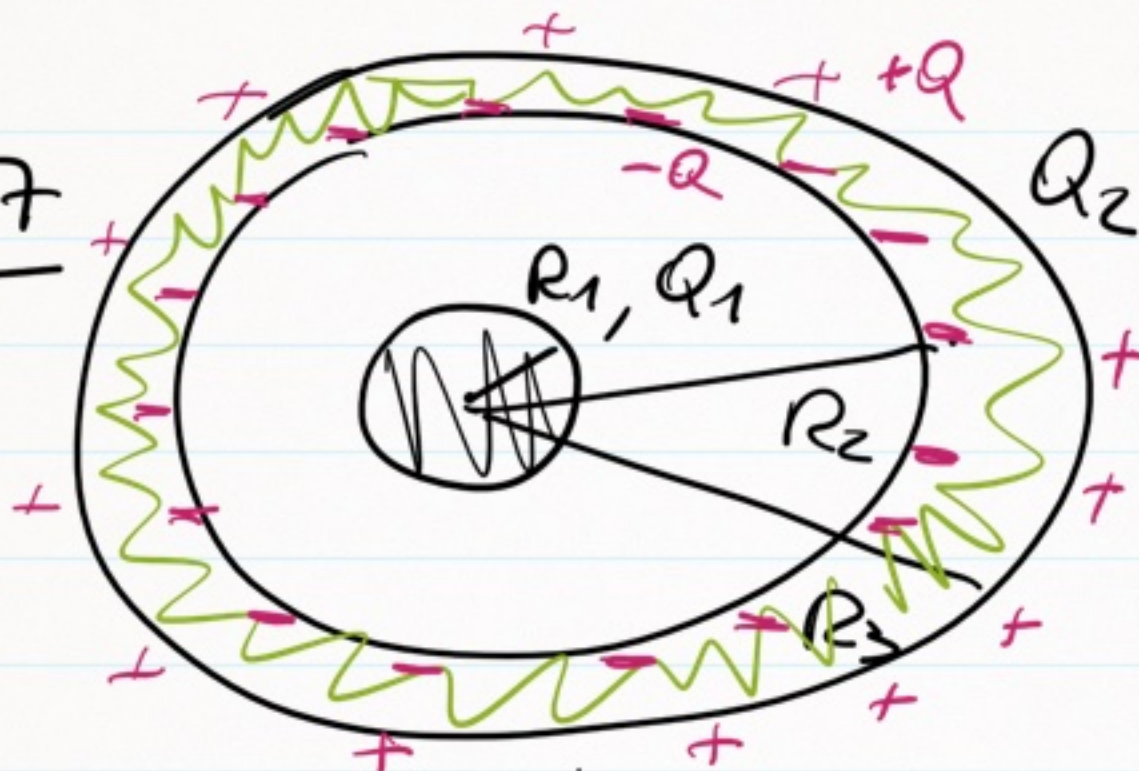
$$V = 899 \text{ V}$$

$$C = \frac{q}{V} = \underbrace{4\pi\epsilon_0 R}_{\text{...}} = 11.12 \text{ pF}$$

$$[C] = \frac{C}{V} = F$$

$$(e) L = \int_0^q V dq = \frac{q^2}{2C} = 449.64 \text{ J}$$

⑥ Ex. 7



(a) $V_0 = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R_1}$

(b) $V_1, V_2, \dots$?

Ricorda: $V(\infty) = 0$

$V_2 = \frac{1}{4\pi\epsilon_0} \frac{Q}{R_3}$

← poi da ∞ mi avvicino,
trovo la superficie di raggio R_3
poi entro all'interno...

$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$

$R_1 < r < R_2$

$$\Delta V_{1-2} = V_1 - V_2 = \int_{R_1}^{R_2} E(r) dr$$

$$= \underbrace{\frac{Q}{4\pi\epsilon_0}}_{V_0 R_1} \int_{R_1}^{R_2} \frac{1}{r^2} dr =$$

$$= V_0 R_1 \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = V_0 \left(1 - \frac{R_1}{R_2} \right)$$

$$\Rightarrow V_1 = V_0 \cdot \frac{R_1}{R_3} + V_0 \left(1 - \frac{R_1}{R_2} \right) =$$

$$= V_0 \left(1 - \frac{R_1}{R_2} + \frac{R_1}{R_3} \right) = \dots$$

$$(c) \quad V_2 = 0 \quad \Rightarrow \quad V_1 = V_0 \left(1 - \frac{R_1}{R_2} \right)$$

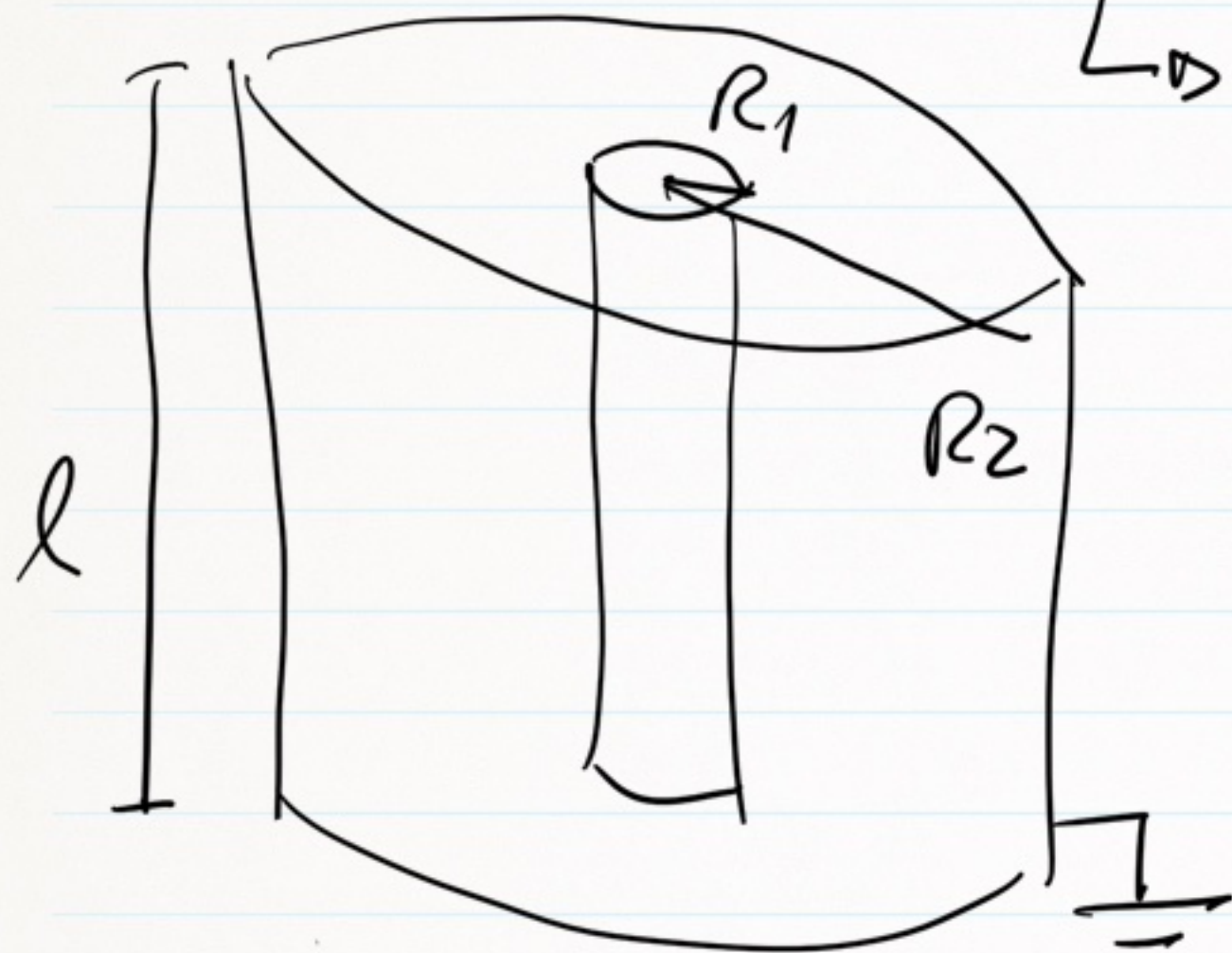
$$(d) \quad V_1 = V_2 = \frac{Q}{4\pi\epsilon_0 R_3}$$

$$C = \frac{Q}{\Delta V_{1-2}} = \frac{4\pi\epsilon_0 R_1 R_2}{R_2 - R_1}$$

⑦ Ex. 10

$l \gg R_1, R_2$

CONDENSATORS
CYLINDRICAL



$$\hookrightarrow E(r) = \frac{1}{2\pi\epsilon_0} \cdot \frac{\Lambda}{r}$$

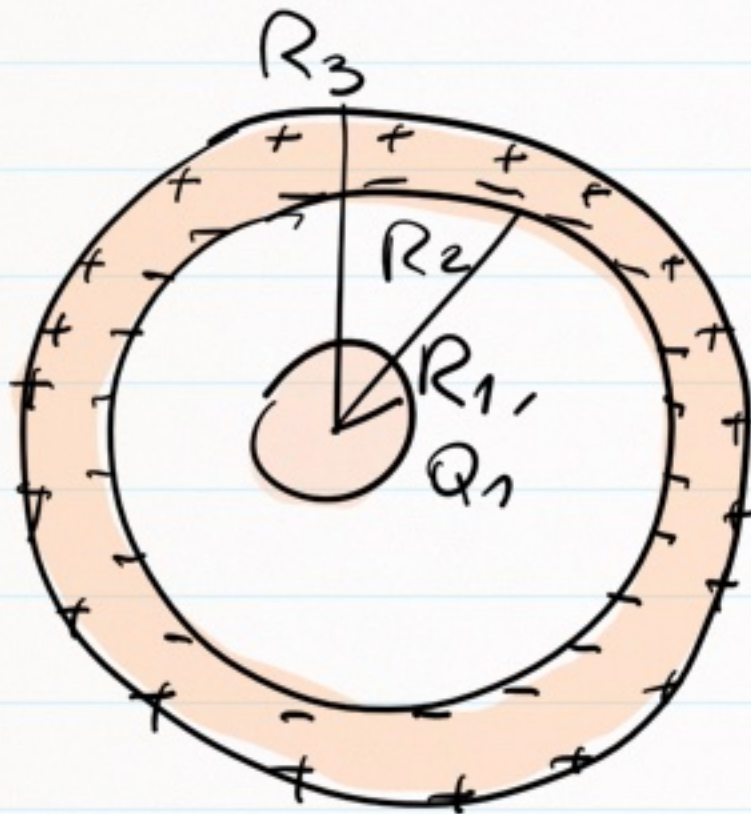
$$= \frac{1}{2\pi\epsilon_0} \frac{Q}{l \cdot r}$$

$$\Delta V_{1-2} = V_1 - V_2 = \int_{R_1}^{R_2} E(r) dr = \frac{Q}{2\pi\epsilon_0 l} \left[\ln r \right]_{R_1}^{R_2}$$

$$= \frac{Q}{2\pi\epsilon_0 l} \cdot \ln \frac{R_2}{R_1}$$

$$C = \frac{Q}{\Delta V_{1-2}} = \frac{2\pi \epsilon_0 l}{\ln(R_2/R_1)}$$

Ex. 8



$$Q_1 = Q$$

uniforme

$$(a) \sigma_2 = -\frac{Q}{4\pi R_2^2} = -\frac{10^{-6} \text{ C}}{4\pi \cdot 10^2 \text{ m}}$$

$$= -7.96 \cdot 10^{-6} \frac{\text{C}}{\text{m}^2}$$

(b) posso usare l'espressione della capacità di un cond. sferico trovato @ Ex. 7(b)

$$\Delta V_{12} = \frac{Q}{C} = \frac{Q}{4\pi \epsilon_0} \cdot \frac{R_2 - R_1}{R_1 R_2} = \frac{10^{-6} \text{ C} \cdot \text{Nm}^2}{4\pi \cdot 8.85 \cdot 10^{-12} \text{ C}^2} \cdot \frac{5 \cdot 10^{-2} \text{ m}}{5 \cdot 10^{-2} \cdot 10^{-1} \text{ m}^2} =$$

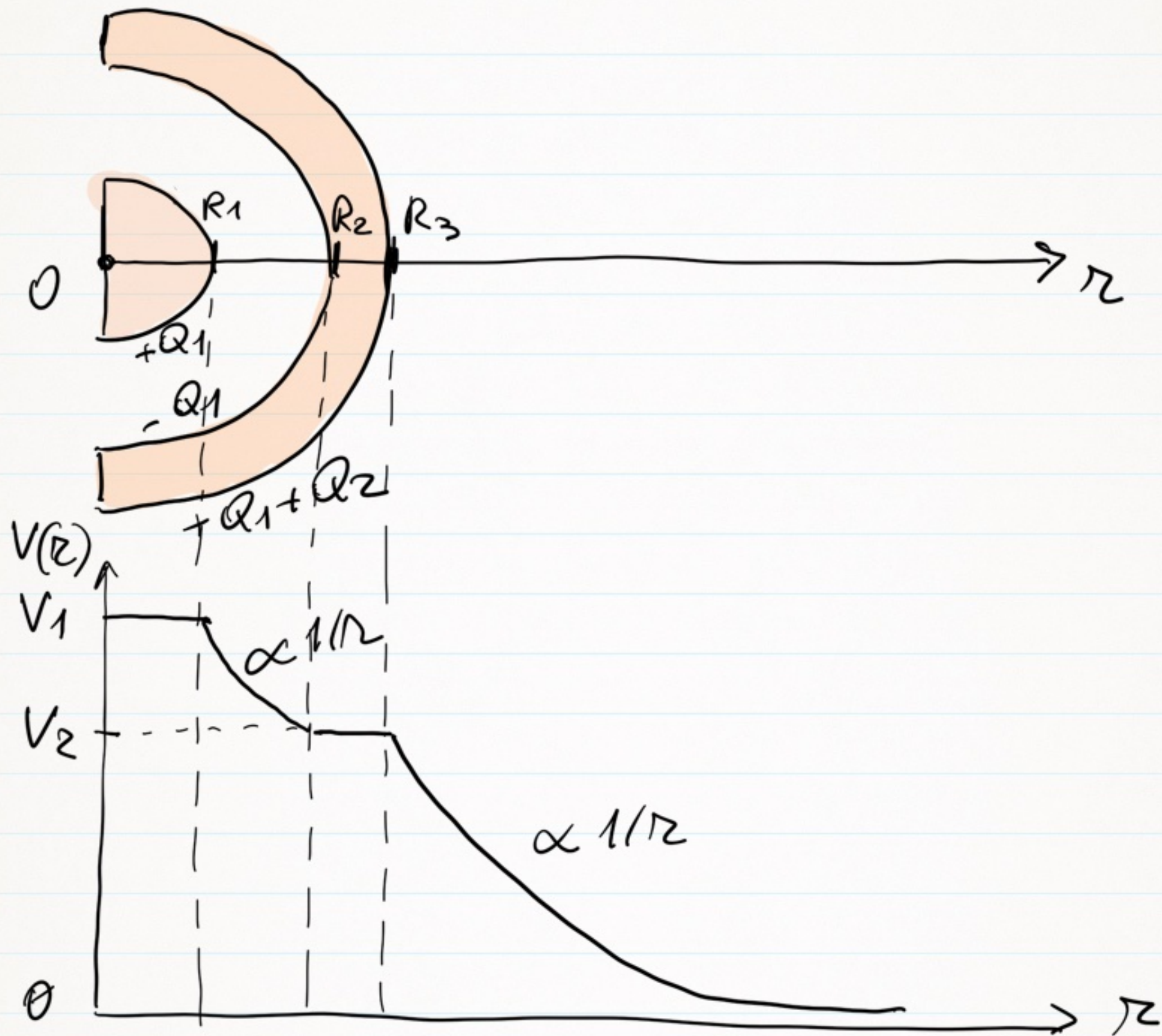
$$\simeq 9 \cdot 10^4 \text{ V}$$

(c) • Come nell'esercizio precedente, posto da ∞ , dove $V = 0$

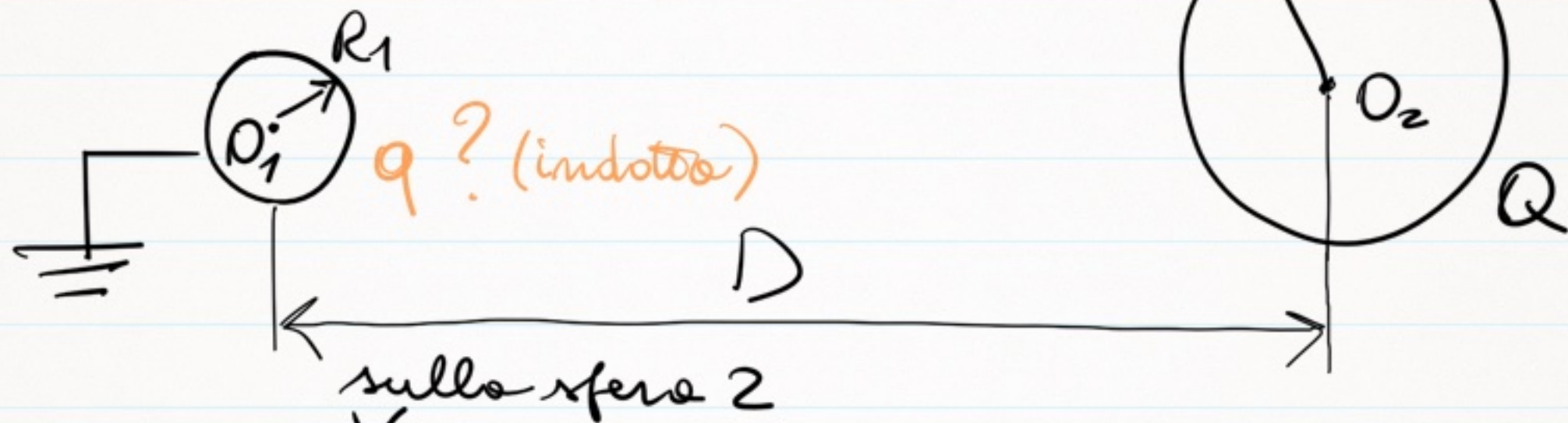
[attenzione: in questo caso, abbiamo visto che per altri problemi non è necessariamente così, dipende dalla distribuzione di carica]

$$\bullet V_2 = \frac{1}{4\pi\epsilon_0} \frac{(Q_1 + Q_2)}{R_3}$$

$$\bullet V_1 = V_2 + \Delta V_{12} = \frac{Q_1 + Q_2}{4\pi\epsilon_0} \cdot \frac{1}{R_3} + \frac{Q_1}{4\pi\epsilon_0} \frac{R_2 - R_1}{R_1 R_2}$$



Esercizio 9



La carica Q induce una redistribuzione della carica sull'altra sferetta (1), producendo una densità sup. non uniforme σ_1 t.c.

$$q = \int_{S_1} \sigma_1 dS_1 \quad \left[\sigma_2 \approx \frac{Q}{4\pi R_2^2} \text{ uniforme} \right]$$

In generale, possiamo quindi scrivere il potenziale nello spazio come somma di 2 contributi -
(SOVRAPPOSIZIONE EFFETTI)

Per trovare 2 incognite posso servirmi di 2 relazioni.

I) r , distanza da O_1

$$V_1(r) = \frac{1}{4\pi\epsilon_0} \int_{S_1} \frac{\sigma_1 dS_1}{r} + \frac{1}{4\pi\epsilon_0} \int_{S_2} \frac{\sigma_2 dS_2}{r}$$

↳ questa relazione traduce la dipendenza del potenziale della sfera 1 dalle sfera 2

II) r' , distanza da O_2

$$V_2(r) = \frac{1}{4\pi\epsilon_0} \int_{S_1} \frac{\sigma_1 dS_1}{r'} + \frac{1}{4\pi\epsilon_0} \int_{S_2} \frac{\sigma_2 dS_2}{r'}$$

↳ traduce la dipendenza del potenziale della sfera 2 dalla sfera 1

I) valutazione sulla sup. di 1

$$V_1 \simeq \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R_1} + \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{D} \simeq 0$$

approx.

perché

$D \gg R_1, R_2$

$$\Rightarrow \boxed{q \simeq - \frac{R_1}{D} \cdot Q}$$

II) valutazione sulla sup. di 2

$$V_2 \simeq \frac{1}{4\pi\epsilon_0} \cdot \underbrace{\frac{q}{D}}_{-\frac{R_1}{D^2}Q} + \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R_2} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R_2} - \frac{R_1}{D^2} \right)$$